

Moritz Schwerer

Reinforcement Learning Researcher | Humanoid Robotics

ROBOTICS PORTFOLIO

moritzschwerer.github.io

Real-world robot demonstrations

Reinforcement learning researcher developing dynamic whole-body skills and real-time teleoperation for humanoid robots using imitation learning, policy distillation, curricula, and robust sim-to-real transfer.

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📍 Lahr, Germany

🐙 github.com/MoritzSchwerer

EDUCATION

M.Sc. Computer Science

Offenburg University of Applied Sciences

10/2024 - 08/2026

Offenburg, Germany

WORK EXPERIENCE

Research Assistant

Offenburg University of Applied Sciences

02/2025 - Present

Offenburg, Germany

- Contributed to Team Sweaty's second-place finish in 3-on-3 soccer at the 2025 World Humanoid Robot Games in Beijing.
- Developed an RL-based whole-body teleoperation controller from AMASS motion data; trained a future-conditioned privileged teacher, distilled it into a causal student, and further fine-tuned the student with RL.
- Built a real-time HTC Vive teleoperation pipeline using Vive Tracker 3.0 end-effector targets streamed via UDP to the onboard tracking policy.
- Developed a teacher-student RL pipeline in PyTorch and Isaac Lab, distilling a privileged controller and pretrained locomotion policy into neural ball-search and approach behaviors.
- Trained command-conditioned dribbling and kicking policies with progressive curricula, building complex soccer skills on top of the distilled approach policy.
- Designed stable runtime skill switching between separately trained policies; used Reference State Initialization to learn robust dribble-to-kick transitions without falls.
- Achieved robust transfer of dynamic skills to physical robots through extensive system identification and domain randomization.
- Integrated policies with game-state planning, team coordination, fallback behaviors, and vision/proprioception localization.
- Optimized the onboard C++ control stack to minimize latency affecting interactions with the ball and opponents.

AI Research Engineer Working Student / Bachelor Thesis

Robert Bosch GmbH

09/2022 - 09/2024

Stuttgart, Germany

- Designed benchmarks comparing internal LiDAR perception networks with state-of-the-art 3D object detectors.
- Implemented and evaluated range-view, voxel-based, and pillar-based LiDAR detectors for far-range and small-object detection.
- Initiated internal R&D on unsupervised pretraining for point-cloud 3D object detectors, enabling substantially larger data and compute budgets than purely supervised training.
- Developed targeted methods for dataset imbalance and improved performance on underrepresented classes.
- Built Azure ML pipelines for reproducible training, evaluation, and comparison of perception models.

TECHNICAL SKILLS

Robot Learning

Reinforcement and imitation learning, policy distillation, curricula, Reference State Initialization, and control.

Sim-to-Real & Deployment

System identification, domain randomization, and deployment and evaluation on physical humanoid robots.

Teleoperation & Human Motion

AMASS motion data, privileged-to-causal policy distillation, HTC Vive tracking, and real-time UDP streaming.

Python & PyTorch

Isaac Lab training pipelines for reinforcement learning, simulation workflows, and policy evaluation.

Autonomous Robotics

Runtime skill switching, game-state planning, team coordination, fallback behaviors, and localization.

C/C++ & Linux

Latency-optimized onboard control, performance-critical robotics components, debugging, and deployment.

Perception & Localization

YOLO-based visual localization and domain-specific fine-tuning; 3D LiDAR detection, pretraining, and Azure ML.

LANGUAGES

German

Native

English

Full professional proficiency